CS182: Introduction to Machine Learning
Lecture 05: Bayesian Decision Theory

Worked Solutions and Marking Guide

Scope: Bayesian_Decision_Theory.pdf. Independent practice material based on the official 2026 Fall lecture.

Total: 100 marks. **Suggested time:** 120 minutes.

Section	Questions	Each	Marks
A. Misconceptions: True/False	12	1	12
B. Multiple response	12	2	24
C. Proofs and formula derivations	4	10	40
D. Applications	2	12	24
Total	30		100

Scoring. In Section B each item has exactly four options, of which one, two, or three are correct. The complete correct set earns 2 marks; a nonempty proper subset earns 1 mark; any incorrect option or no selection earns 0. T/F answers earn 1 mark only when correct. Show reasoning in Sections C and D; part marks are available for valid intermediate work. Equivalent correct methods receive credit.

Instructions. Answer in English. Use natural logarithms. Exact values and justified approximations are accepted. All vectors are columns unless stated otherwise. Use local assumptions, label conventions, and tie rules.

Quick key — Section A

1	2	3	4	5	6
F	T	F	T	T	F

7	8	9	10	11	12
F	F	F	T	T	F

Quick key — Section B

13	14	15	16	17	18
BCD	ACD	BCD	CD	ABC	ABD

19	20	21	22	23	24
A	AD	AC	ABC	BC	ABC

Section A. True/False — Explanations

1. False. A continuous density can exceed 1; its integral, rather than its height, is normalized. Class posterior probabilities $P(C_i \mid x)$ do lie in $[0, 1]$.
Source: PDF pp.11-12.

2. True. For example, if $P(C_1 \mid x) = 0.3$, the loss of predicting C_1 when C_2 is true is 1, and the opposite error costs 4, then $R(\alpha_1 \mid x) = 0.7 < 1.2 = R(\alpha_2 \mid x)$.
Source: PDF p.15; supplementary numerical illustration.

3. False. The tied scores must also be largest. If another class has a strictly larger score, the equality is hidden inside that class's decision region.
Source: PDF p.23.

4. True. Every pairwise difference is multiplied by the same positive number $a(x)$. Its sign and any zero difference are preserved pointwise.
Source: PDF p.21; supplementary pointwise extension.

5. True. The error of choosing C_i is $1 - P(C_i)$; minimizing it is equivalent to maximizing the prior. Ties allow several optimal constant decisions.
Source: PDF pp.6-7,16.

6. False. Within-class monotonicity does not preserve comparisons across classes. Scores $g_1 = 2, g_2 = 1$ choose C_1, whereas $f_1(t) = t$ and $f_2(t) = t + 10$ choose C_2. The same increasing function must be used for all classes.
Source: PDF p.21.

7. False. For a fixed x, the same positive denominator does not change the maximizing class. It is still required to make $\sum_i P(C_i \mid x) = 1$.
Source: PDF pp.11-12,21.

8. False. The relevant scores are $p(x \mid C_i)P(C_i)$. Maximum likelihood classification agrees automatically with MAP when the priors are equal; unequal priors can reverse the ordering.
Source: PDF pp.11-12,16.

9. False. The mixed risk is a weighted average of the action risks and cannot be below their minimum. Randomization among minimizers can tie that minimum.
Source: PDF p.15; supplementary randomization argument.

10. True. Their risks are both $1 - \max_k P(C_k \mid x)$. A tie-breaking convention determines the output label without changing this minimum risk.
Source: PDF pp.12,16.

11. True. The best classification risk is $1 - \max_i P(C_i \mid x)$, while rejection has risk λ. The strict comparison gives the stated threshold; equality is a risk tie.
Source: PDF pp.17-18.

12. False. The required factorization is $p(x \mid C_i) = \prod_j p(x_j \mid C_i)$, separately within each class. Mixing classes can produce dependent features even if this conditional factorization holds.
Source: PDF p.13.

Section B. Multiple Response — Explanations

13. Correct set: B, C, D.

(A) **Incorrect.** Its score is always strictly smaller than g_3.

(B) **Correct.** It is $x < -1/\sqrt{2}$ or $x > 1/\sqrt{2}$.

(C) **Correct.** These solve $g_1 = g_3 = 1/2$, and those tied scores exceed $g_2 = 0$.

(D) **Correct.** At both points $g_3 = 1/2$ exceeds the tied value 0.

Source: PDF p.23; supplementary algebraic example.

14. Correct set: A, C, D.

(A) **Correct.** The constant rule is available among all rules using X, so optimization cannot have a larger risk.

(B) **Incorrect.** A feature-dependent rule may ignore useful posterior information or deliberately choose a low-posterior class.

(C) **Correct.** The pointwise minimum error $1 - \max_i P(C_i \mid X)$ is zero almost surely.

(D) **Correct.** Then $E[1 - \max_i P(C_i \mid X)] = 1 - \max_i P(C_i)$.

Source: PDF pp.7,9-12,15-16; supplementary comparison of rule classes.

15. Correct set: B, C, D.

(A) **Incorrect.** The acceptance threshold is $1 - 0.55 = 0.45$, not 0.55.

(B) **Correct.** The best class risk 0.54 already exceeds 0.50.

(C) **Correct.** The best classification risk is $1 - 0.46 = 0.54 < 0.55$.

(D) **Correct.** Rejection is strictly optimal when $1 - \max_i P(C_i \mid x) > \lambda$; increasing λ makes this condition harder to satisfy.

Source: PDF pp.17-18.

16. Correct set: C, D.

(A) **Incorrect.** A negative multiplier reverses strict order, making maximizers into minimizers.

(B) **Incorrect.** Class-dependent offsets can reverse cross-class comparisons.

(C) **Correct.** Strict monotonicity preserves both strict order and ties across all classes.

(D) **Correct.** The pairwise differences become $a(x)[g_i(x) - g_j(x)]$, preserving their signs.

Source: PDF p.21; supplementary pointwise extension.

Section B. Multiple Response — Explanations

17. Correct set: A, B, C.

(A) **Correct.** Every action risk gains the same $\sum_k b_k P(C_k \mid x)$.

(B) **Correct.** Maximizing $-R(\alpha_i \mid x)$ is equivalent to minimizing $R(\alpha_i \mid x)$.

(C) **Correct.** All action risks are multiplied by that constant, preserving order and ties.

(D) **Incorrect.** Each action then gains a different offset, which can change the minimizing action.

Source: PDF pp.15,20; supplementary loss transformations.

18. Correct set: A, B, D.

(A) **Correct.** The log posterior odds equal $\log L(x) + \log(1/3)$.

(B) **Correct.** The joint-score comparison is $L(x) > P(C_2)/P(C_1) = 3$.

(C) **Incorrect.** It is the same positive denominator for both posteriors and cancels.

(D) **Correct.** The threshold is the prior odds $0.9/0.1$.

Source: PDF pp.11-12,21-22.

19. Correct set: A.

(A) **Correct.** This is exactly the factorization $p(x \mid C_i) = \prod_j p(x_j \mid C_i)$.

(B) **Incorrect.** Conditioning can introduce dependence; independence in the mixture does not establish independence within each class.

(C) **Incorrect.** Bayes scores remain $P(C_i) \prod_j p(x_j \mid C_i)$, so unequal priors still matter.

(D) **Incorrect.** Naive Bayes factors feature likelihoods, not posteriors of the class. Repeated posterior factors also repeatedly include the prior.

Source: PDF pp.11-13.

20. Correct set: A, D.

(A) **Correct.** The posterior vector is $(0.45, 0.30, 0.25)$, so C_1 has the largest posterior.

(B) **Incorrect.** The maximum posterior is 0.45; the error is its complement, 0.55.

(C) **Incorrect.** It is $0.18/0.40 = 0.45$; the likelihood alone is not the posterior.

(D) **Correct.** Summing the joint scores gives $0.18 + 0.12 + 0.10 = 0.40$.

Source: PDF pp.11-12,16.

Section B. Multiple Response — Explanations

21. Correct set: A, C.

(A) **Correct.** A convex combination of their equal risks remains 0.6.

(B) **Incorrect.** Its risk is $1 - 0.2 = 0.8$. The rejection threshold is $1 - 0.6 = 0.4$, not 0.6.

(C) **Correct.** Predicting either of the first two classes costs 0.6, as does rejection.

(D) **Incorrect.** The two largest-posterior class actions tie with rejection. A reject-on-equality convention selects one of several minimizers.

Source: PDF pp.15-18; supplementary tie/randomization interpretation.

22. Correct set: A, B, C.

(A) **Correct.** This error occurs when C_2 is true, giving $8(1 - 0.55) = 3.6$.

(B) **Correct.** Its risk is $2(0.55) = 1.1 < 3.6$.

(C) **Correct.** The inequality $8(1 - q) < 2q$ is equivalent to $q > 0.8$.

(D) **Incorrect.** Posterior maximization minimizes unit error loss; these asymmetric costs produce a different threshold.

Source: PDF pp.15-16; supplementary asymmetric-loss specialization.

23. Correct set: B, C.

(A) **Incorrect.** The mixed risk is $0.2 + 0.3w_2$, which is at least 0.2.

(B) **Correct.** Equality at the minimum requires $w_2 = 0$, because its risk is strictly larger.

(C) **Correct.** Its expected risk is $(0.2 + 0.2)/2 = 0.2$.

(D) **Incorrect.** Any mixture supported on the first and third actions is optimal.

Source: PDF p.15; supplementary randomization derivation.

24. Correct set: A, B, C.

(A) **Correct.** It is $4(0.1) + 1(0.9) = 1.3$; a density value may exceed 1.

(B) **Correct.** It is $0.4/1.3 = 4/13$.

(C) **Correct.** Its joint score 0.9 exceeds 0.4, so its posterior is $9/13$.

(D) **Incorrect.** The prior-weighted likelihoods determine the posterior ordering.

Source: PDF pp.11-12,16.

Section C. Proofs and Formula Derivations

25. Why pointwise decisions minimize total risk **[10 marks]**

(a) [4 marks] Conditioning first on X gives

$$\mathcal{R}(\delta) = E_X\left[\sum_k \lambda_{\delta(X),k} P(C_k \mid X)\right] = \int R(\delta(x) \mid x) p(x)\, dx.$$

Use a sum instead of an integral for discrete X. For a measurable minimizing selector δ^*, $R(\delta^*(x) \mid x) \le R(\delta(x) \mid x)$ wherever posteriors are defined. Multiplication by $p(x) \ge 0$ and integration prove $\mathcal{R}(\delta^*) \le \mathcal{R}(\delta)$. Choices on sets of X-probability zero do not affect total risk; ties can be resolved arbitrarily. Marking: conditional expectation (2), pointwise inequality and integration (1), ties/null sets (1).

(b) [3 marks] For action α_i,

$$R(\alpha_i \mid x) = \sum_{k \ne i} P(C_k \mid x) = 1 - P(C_i \mid x).$$

Thus any largest-posterior class is a conditional-risk minimizer. Substitution into the total-risk formula yields

$$\mathcal{R}^* = E_X[1 - \max_i P(C_i \mid X)].$$

Marking: risk simplification (1), MAP minimizer (1), expected minimum error (1).

(c) [3 marks] Choose $j \in \arg\max_i P(C_i)$. Pointwise $\max_i P(C_i \mid X) \ge P(C_j \mid X)$. Taking expectations and using the law of total probability gives $E[\max_i P(C_i \mid X)] \ge E[P(C_j \mid X)] = P(C_j)$. Subtracting from 1 proves the bound. The result concerns the true Bayes rule; an estimated or poorly chosen classifier need not minimize true risk. Marking: pointwise bound (1), expectation and final inequality (1), scope of guarantee (1).

Marking: use the displayed subpart allocations; award credit for the logical or numerical intermediate steps.

Source: PDF pp.15-16; supplementary integration and information comparison.

Section C. Proofs and Formula Derivations

26. Asymmetric errors and rejection thresholds [10 marks]

(a) [4 marks] The risks are $R_1 = a(1-q)$ and $R_2 = bq$. Therefore $R_1 < R_2$ exactly when $q > a/(a+b)$; reverse the inequality for C_2, with a tie at equality. Equivalently $q/(1-q) > a/b$. Bayes' rule gives

$$\frac{q}{1-q} = \frac{p(x \mid C_1)P(C_1)}{p(x \mid C_2)P(C_2)},$$

so predict C_1 when

$$\frac{p(x \mid C_1)}{p(x \mid C_2)} > \frac{aP(C_2)}{bP(C_1)}.$$

Marking: risks (1), posterior threshold and tie (1), posterior odds (1), likelihood-ratio threshold (1).

(b) [3 marks] Rejection is strictly optimal precisely when both $r < a(1-q)$ and $r < bq$, hence

$$\frac{r}{b} < q < 1 - \frac{r}{a}.$$

This interval has positive width exactly when $r/b < 1 - r/a$, or $r < ab/(a+b)$. At either endpoint rejection ties a class action; at $r = ab/(a+b)$ it only ties at the original class threshold. For larger r, rejection is never strictly optimal. Marking: two risk comparisons (1), interval (1), width condition and ties (1).

(c) [3 marks] The best class risk is $1 - M(x)$, where $M(x) = \max_i P(C_i \mid x)$, and reject risk is λ. Classification is strictly preferable exactly when $M(x) > 1 - \lambda$; rejection is strictly preferable when $M(x) < 1-\lambda$. Since the K posteriors sum to 1, $M(x) \geq 1/K$. If $\lambda > 1-1/K$, then $1-\lambda < 1/K \leq M(x)$, so a class action strictly beats rejection everywhere. At equality of the costs, the uniform posterior can produce a tie. Marking: risks (1), threshold/tie distinction (1), $1/K$ bound and consequence (1).

Marking: use the displayed subpart allocations; award credit for the logical or numerical intermediate steps.

Source: PDF pp.15-18; supplementary asymmetric-cost extension.

Section C. Proofs and Formula Derivations

27. Equivalent discriminants and the role of strict monotonicity [10 marks]

(a) [4 marks] For any two scores u, v, strict monotonicity gives $u > v \iff f(u) > f(v)$, and injectivity gives $u = v \iff f(u) = f(v)$. Thus a class is at least as large as every competitor before transformation exactly when it is afterward. The full argmax set is unchanged. If f is merely nondecreasing, it can merge different scores: take $g_1 = 2, g_2 = 1$ and $f(t) = 0$ for all t. The original argmax is only C_1, whereas both classes maximize transformed scores. Marking: order/tie argument (2), argmax conclusion (1), counterexample (1).

(b) [3 marks] Bayes' rule gives $P(C_i \mid x) = p(x \mid C_i)P(C_i)/p(x)$. Multiplying all scores by common $p(x) > 0$ preserves their ordering and ties, so the posterior and joint scores have identical argmax sets. Applying the common strictly increasing logarithm yields $\log p(x \mid C_i) + \log P(C_i)$, with the same maximizers. If a joint score is zero, the finite logarithm is undefined; one may compare the original scores or use the extended value $\log 0 = -\infty$. At least one score is positive when $p(x) > 0$. Marking: common denominator (1), log decomposition (1), zero-score qualification (1).

(c) [3 marks] Subtract the two log joint scores:

$$g(x) = \log \frac{p(x \mid C_1)}{p(x \mid C_2)} + \log \frac{P(C_1)}{P(C_2)}.$$

If $g > 0$, the posterior of C_1 is larger; if $g < 0$, that of C_2 is larger; $g = 0$ is a tie. For arbitrary losses, the optimal action minimizes $\sum_k \lambda_{ik} P(C_k \mid x)$, which need not be the maximum-posterior class. Scores $h_i(x) = -R(\alpha_i \mid x)$ give the correct maximizing action. Marking: log-ratio derivation (1), sign/tie rule (1), general-loss argument (1).

Marking: use the displayed subpart allocations; award credit for the logical or numerical intermediate steps.

Source: PDF pp.20-22; supplementary tie-preserving conditions.

Section C. Proofs and Formula Derivations

28. What a decision boundary actually represents **[10 marks]**

(a) [4 marks] Each comparison is

$$(w_i - w_j)^\mathsf{T} x + (b_i - b_j) \geq 0,$$

a closed half-space (possibly the whole space or empty). Thus $\mathcal{R}_i$ is an intersection of half-spaces. If u, v satisfy every inequality, then for $0 \leq t \leq 1$ each left side at $tu + (1-t)v$ is the corresponding convex combination of nonnegative values, so the segment also lies in $\mathcal{R}_i$. Equality $g_i = g_j$ is a hyperplane when $w_i \neq w_j$, but only its portions where these scores are maximal can be decision boundaries. Marking: inequalities (1), intersection/segment proof (2), equality and maximality condition (1).

(b) [3 marks] Class C_1 wins strictly when $x^2 > 4$, that is, $x < -2$ or $x > 2$. These are two disconnected components. Class C_2 wins strictly when $-2 < x < 2$. At $x = \pm 2$, the two scores tie for the maximum and form the boundaries. The weak regions include the endpoints, and any tie convention assigns them without affecting the disconnected strict regions. Marking: exterior region (1), interior and boundaries (1), disconnectedness/ties (1).

(c) [3 marks] The equality $g_1 = g_2$ gives $x = \pm\sqrt{2}$, while $g_1 = g_3$ gives $x = \pm 1$; $g_2 = g_3$ has no solution. At $\pm\sqrt{2}$, the tied first two scores are 0 but $g_3 = 1$, so these points lie strictly in C_3's region. At ± 1, $g_1 = g_3 = 1 > 0 = g_2$, and the winner changes between C_1 and C_3. The only boundaries are ± 1. Marking: pairwise equalities (1), largest-score check (1), actual boundaries and counterexample conclusion (1).

Marking: use the displayed subpart allocations; award credit for the logical or numerical intermediate steps.

Source: PDF p.23; supplementary affine and quadratic examples.

Section D. Applications

29. A sensor with asymmetric mistakes and a review action **[12 marks]**

(a) [4 marks] The two joint probabilities at A are $0.175, 0.065$; at B, $0.105, 0.260$; at C, $0.070, 0.325$. Therefore

$$P(A) = 0.240, \quad P(B) = 0.365, \quad P(C) = 0.395,$$

which sum to 1. The posteriors are

$$q_A = \frac{35}{48} \approx 0.729167, \quad q_B = \frac{21}{73} \approx 0.287671, \quad q_C = \frac{14}{79} \approx 0.177215.$$

Marking: joint scores (1), report probabilities (1), posterior values (2).

(b) [4 marks] For posterior q, the risks are $3(1-q)$, $9q$, and 1.7. At A, these are $13/16 = 0.8125$, $105/16 = 6.5625$, and 1.7, so predict C_1. At B, they are $156/73 \approx 2.136986$, $189/73 \approx 2.589041$, and 1.7, so review. At C, they are $195/79 \approx 2.468354$, $126/79 \approx 1.594937$, and 1.7, so predict C_2. Equivalently, review is preferred for $17/90 < q < 13/30$. Marking: risk formulas (1), three risk triples with correct actions (3).

(c) [4 marks] The review-enabled expected loss is

$$0.240(0.8125) + 0.365(1.7) + 0.395\left(\frac{126}{79}\right) = 0.195 + 0.6205 + 0.630 = 1.4455.$$

Without review, the posterior threshold is $3/(3+9) = 0.25$, giving C_1 at A, B and C_2 at C. Its loss is $3(0.065) + 3(0.260) + 9(0.070) = 1.605$. MAP uses threshold 0.5, giving C_1 only at A; its loss is $3(0.065) + 9(0.105) + 9(0.070) = 1.770$. Review reduces the forced minimum loss by 0.1595. Marking: review expectation (2), forced-cost rule/loss (1), MAP rule/loss and comparison (1).

Marking: use the displayed subpart allocations; award credit for the logical or numerical intermediate steps.

Source: PDF pp.11-12,15-18; supplementary asymmetric review-cost application.

Section D. Applications

30. Conditional independence, posterior decisions, and rejection [12 marks]

(a) [4 marks] The joint-score pairs $(P(x, C_1), P(x, C_2))$ are

$$00 : (0.056, 0.135), \quad 01 : (0.024, 0.315), \quad 10 : (0.224, 0.045), \quad 11 : (0.096, 0.105).$$

For example, the C_1 score at 10 is $0.4(0.8)(0.7) = 0.224$. The winning classes are C_2, C_2, C_1, C_2, respectively. Summing each losing joint score gives the error

$$0.056 + 0.024 + 0.045 + 0.096 = 0.221.$$

Marking: four joint-score pairs (2), decisions (1), error sum (1).

(b) [4 marks] The feature-pair probabilities are $0.191, 0.339, 0.269, 0.201$. Their maximum posteriors are

$$\frac{0.135}{0.191} \approx 0.706806, \quad \frac{0.315}{0.339} \approx 0.929204, \quad \frac{0.224}{0.269} \approx 0.832714, \quad \frac{0.105}{0.201} \approx 0.522388.$$

The acceptance threshold is $1 - 0.30 = 0.70$. Accept $00, 01, 10$ with the previous MAP labels; reject 11. Total risk is

$$0.056 + 0.024 + 0.045 + 0.30(0.201) = 0.1853.$$

This includes review cost on all rejected inputs rather than treating them as free. Marking: posterior maxima/threshold (2), accepted/rejected pairs (1), expected risk (1).

(c) [4 marks] The marginal probabilities are

$$P(X_1 = 1) = 0.4(0.8) + 0.6(0.25) = 0.47, \quad P(X_2 = 1) = 0.4(0.3) + 0.6(0.7) = 0.54.$$

Their joint probability is $P(11) = 0.096 + 0.105 = 0.201$, whereas $0.47(0.54) = 0.2538$. Since these differ, the features are dependent marginally. This is compatible with the stated conditional independence: both features respond to the shared, unobserved class variable. Marking: marginals (2), joint-versus-product comparison (1), conditional/marginal distinction (1).

Marking: use the displayed subpart allocations; award credit for the logical or numerical intermediate steps.

Source: PDF pp.11-13,16-18; supplementary marginal-dependence calculation.

Lecture Alignment and Coverage

Primary source: `Bayesian_Decision_Theory.pdf` in `E:\CS182\2026fall\lecture`. All PDF pages count the cover as page 1. Additional arguments are marked as supplementary derivations when appropriate. These are original practice questions, not copied homework questions.

T/F

1 PDF pp.11-12.

2 PDF p.15; supplementary numerical illustration.

3 PDF p.23.

4 PDF p.21; supplementary pointwise extension.

5 PDF pp.6-7,16.

6 PDF p.21.

7 PDF pp.11-12,21.

8 PDF pp.11-12,16.

9 PDF p.15; supplementary randomization argument.

10 PDF pp.12,16.

11 PDF pp.17-18.

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Multiple response

13 PDF p.23; supplementary algebraic example.

14 PDF pp.7,9-12,15-16; supplementary comparison of rule classes.

15 PDF pp.17-18.

16 PDF p.21; supplementary pointwise extension.

17 PDF pp.15,20; supplementary loss transformations.

18 PDF pp.11-12,21-22.

19 PDF pp.11-13.

20 PDF pp.11-12,16.

21 PDF pp.15-18; supplementary tie/randomization interpretation.

22 PDF pp.15-16; supplementary asymmetric-loss specialization.

23 PDF p.15; supplementary randomization derivation.

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